**ISYE 7406**

# Public Health Operational Efficiency: Predicting Physical Inactivity in Texas and California Adults

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## ABSTRACT

Physical inactivity is a primary clinical driver of chronic health conditions, yet regional and socioeconomic determinants often remain obscured in aggregate national data. This study presents a predictive modeling framework using the 2022 Behavioral Risk Factor Surveillance System to identify key determinants of physical inactivity among adults in California and Texas. Utilizing a train, validation and test split, a Baseline Logistic Regression, Stepwise Selection , Ridge Regression, and LASSO methods were utilized to understand physical inactivity. Also, the regularized models were optimized using a 10 fold cross-validation. This produced strong predictive performance, with test accuracy of 78.35% and a ROC-AUC of 0.7431, and the four models performing almost identically. Self-reported general health, age, and household income emerged as the most salient risk drivers across both states, demonstrating that socioeconomic gradients transcend state boundaries. These results offer actionable insights for public health entities to target physical inactivity interventions in those states.

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## 1. INTRODUCTION

Physical inactivity is widely recognized as a major public health challenge in the United States, contributing to increased rates of cardiovascular disease, type 2 diabetes, obesity, and other types of disorders. Given that the United States operates under a federal system in which resource allocation and public policy implementation are largely decentralized, an important question arises: does physical inactivity follow similar patterns in every state?

Texas and California are currently the two most populous states in the country. Likewise, according to the most recent state economic performance data, they also have the highest nominal Gross Domestic Product (GDP) among all U.S. states. Given that the United States operates under a federal system in which resource allocation and public policy implementation are largely decentralized, an important question arises: does physical inactivity follow similar patterns in these two states? Understanding this is particularly relevant because it can provide valuable insights into the demographic, socioeconomic, and clinical factors associated with physical inactivity in different regional contexts.

A better understanding of these factors can help inform more effective health promotion strategies, regulatory initiatives, and public policy interventions tailored to the specific characteristics and needs of each state. By identifying how physical inactivity differ between Texas and California, policymakers and public health professionals can design more targeted and impactful interventions to improve population health.

Accordingly, the research was structured into a three-phase framework, allowing us to work with messy, real world, high dimensional health data. Which is consisting of the following stages:

**Project Workflow Overview**

- **Phase 1:** Data Preprocessing and Exploration
    - The project begins with cleaning and standardizing numeric and categorical variables to ensure data quality.
    - Missing values and nonresponse fields common in large health surveys are addressed through imputation.
    - Visual exploratory data analysis (EDA) is performed to analyze correlations between factors such as income, BMI, and geographic location to detect regional patterns.

- **Phase 2:** Model Training and Optimization
    - The dataset is partitioned into training, validation, and testing sets to benchmark performance and mitigate overfitting.
    - Multiple models are implemented in R, including a Logistic Regression baseline, Stepwise Selection (stepAIC), and regularized models like Ridge and LASSO.
    - The team will utilize k-fold cross validation via cv.glmnet to tune and identify the optimal penalty parameters (beta) for the regularized models.

- **Phase 3:** Evaluation and Strategic Application
    - Performance is assessed using metrics that account for class imbalance, specifically prioritizing Recall and ROC-AUC over raw accuracy.
    - The team will interpret findings by extracting coefficient pathways and variable selection charts to answer the research questions.
    - Finally, the results will be translated into actionable, operational strategies for public health administrators to design targeted regional interventions.

## 2. PROBLEM STATEMENT & DATA SOURCES

### 2.1 Problem Statement

**Physical inactivity is a known clinical driver for chronic disease. Unfortunately, many public policy solutions come from broad quantitative analyses that can miss socioeconomic and behavioral drivers at the state level. Probabilistic reasoning is a critical element missing from much public health analysis. While there are datasets at the** macro level, we lack finely tuned probability models that can help assess risk drivers within localized regions of highly populated states like Texas and California.

Absent this detailed information, those tasked with policy implementation are often unable to take appropriately targeted actions. This study aims to bridge this divide. Rather than simply describing correlates of physical inactivity at a summary level, we leverage the power of machine learning to identify determinants of inactivity at the state level in Texas and California. To that end, this project addresses five key research questions:

1. **What are the most** influential demographic and **clinical predictors of** leisure-time physical inactivity?

2. How significantly do socioeconomic factors (e.g., household income and education levels) impact physical inactivity probability?

3. Does geographic location (California vs. Texas) introduce significant regional baseline differences in physical activity patterns after adjusting for individual covariates?

4. How do regularized classification techniques (Ridge and LASSO) perform relative to traditional baseline logistic regression and stepwise feature selection?

5. How can decision thresholds be calibrated to maximize the identification of at-risk inactive individuals for public health allocation?

### 2.2 Data Sources

This study used the 2022 Behavioral Risk Factor Surveillance System **(BRFSS) dataset as its primary data source. The BRFSS is an annual health survey conducted by the Centers for Disease Control and Prevention** and was administered across all 50 states and dependencies of United States. **The BRFSS collects self reported information from adults through both landline and cellular telephone interviews. The survey gathers data on health-related behaviors, chronic health conditions, access to healthcare services, and the use of preventive health services**.

For this study, data from California and Texas were extracted from the original BRFSS dataset. California was identified by _STATE = 6, while Texas was identified by _STATE = 48. Different variables were selected based on the study's methodological framework. These variables represent a combination of socioeconomic, demographic, and health-related factors that were considered relevant.

The Y variable in this study was physical inactivity, which is identified in the BRFSS dataset as _TOTINDA. This was binary variable, where 0 indicates that the respondent is physically active and 1 indicates physical inactivity. After data cleaning, the final dataset included 21,415 observations. Table 1 provides a detailed description of the variables included in the analysis and their corresponding characteristics.

| Variable Name | BRFSS Variable | Data Type | Description / Categories |
|---|---|---|---|
| **inactive** | _TOTINDA | Binary (Target) | 0 = Physically active during the past 30 days; 1 = Physically inactive during the past 30 days |
| **age** | _AGE80 | Numeric | Age in years (18–80 years) |
| **bmi** | _BMI5 | Numeric | Body Mass Index |
| **genhlth** | GENHLTH | Ordinal | Excellent, Very good, Good, Fair, Poor |
| **income** | INCOME3 | Ordinal | Under 25k, 25k–50k, 50k–75k, 75k–100k, 100k+ |
| **educ** | EDUCA | Ordinal | Less than High School, High School Graduate, Some College, College Graduate |
| **diabetes** | DIABETE4 | Categorical | No, Prediabetes, Yes |
| **sex** | SEXVAR | Binary | Male, Female |
| **state** | _STATE | Binary Factor | California, Texas |

***Table 1. Variables of Study and its characteristics***

## 3. PROPOSED METHODOLOGY

### 3.1 Data Preparation

Before implementing and comparing the four modeling approaches described in the introduction, the BRFSS dataset underwent a data preparation and cleaning process. This step was necessary to facilitate data handling and ensure that the variables were in a suitable for the analysis of the code implementation. As part of this process, the nine variables selected for the analysis were renamed using shorter aliases to simplify the coding procedure. The new variable

names are listed in Table 1. In addition, the BMI variable was converted back to its original scale by dividing the recorded values by 100.

During the initial exploration of the original dataset, missing values were identified, particularly for the INCOME3 variable. Rather than excluding these observations, a new category labeled "Not Reported" was created to retain as much information as possible and avoid potential bias associated with removing individuals who may have been either physically active or inactive. In contrast, observations with missing values (NA)for other numerical variables, such as BMI, were excluded because a reliable value could not be derived for those cases. After completing the data cleaning process, the final analytical sample consisted of 21,415 observations. For this study R was the principal language of work.

### 3.2 Data Separation

To evaluate the performance of the proposed models (Logistic Regression, Stepwise Regression, Ridge, and LASSO) the cleaned dataset was divided into three subsets. Seventy percent of the data were allocated to the training set, while the remaining observations were split evenly between a validation set (15%) and a test set (15%). This data partitioning allowed the models to be trained on a representative sample while reserving separate datasets for model tuning and the final performance evaluation.

### 3.3 Models

The models utilized in this study were the following:

1. **Ridge Regression:** This model applied an L2 penalty on the coefficients to reduce multicollinearity. Ridge keeps all of the predictors in the model but shrinks the magnitude of their coefficients to improve the stability of the estimates. As with the LASSO model, the optimal regularization parameter was selected using 10-fold cross-validation through the cv.glmnet function in R.
2. **LASSO**: In this model, we applied **an L1 penalty that can** shrink non-**essential coefficients to** exactly zero. As a result, LASSO performs variable selection and coefficient shrinkage simultaneously, making it particularly useful **given the number of** predictors included in the **filtered dataset**. As with the Ridge model, the optimal regularization parameter was selected using 10-fold cross-validation through the cv.glmnet function in R.
3. **Logistic Regression**: This model included **all candidate predictors. This** served **as reference point for classification performance and gives direct odds ratio interpretations for each variable, which are straightforward to communicate to a non-technical audience**.
4. Stepwise: Using the Akaike Information Criterion, the model ran both forward and backward selection paths so the algorithm added and dropped predictors based on model fit penalized for complexity, which trims noise variables and isolates a smaller set

that still explains the outcome well. **This directly addresses the high dimensionality of the dataset and produces a model that is easier to interpret and act on**.

## 4. ANALYSIS AND RESULTS

### 4.1 Phase 1: Data Preprocessing and Exploration (EDA)

After filtering out missing data and getting our California (CA) and Texas (TX) sample cleaned and filtered, some preliminary exploration on our dataset was performed to assess feature distributions, multicollinearity amongst features, as well as intuition around risk factors for Leisure time Physical Inactivity:

- Distribution of Age with Physical Inactivity Levels: Comparing the age of respondents between active and inactive individuals shows a stark increase in age towards older ages for those who are physically inactive. While the median age of physically active respondents is around 51 years old, the median age of physically inactive respondents is around 59 years old. In addition, if we compare the interquartile range of these two populations, inactive respondents are generally older (approximately 44 to 72 years old) than physically active respondents (ages 35 to 66). Age is shown to be a crucial demographic factor when it comes to risk of physical inactivity.
- Correlation Check (bmi n age): To ensure there was no collinearity between our numeric predictors before moving onto our regularized regression models, We created a correlation matrix between body mass index (bmi) and age (age). We see that the correlation value is .02, which is essentially no linear relationship. Therefore, bmi and age provide separate, orthogonal sources of information to the model and there is no need to worry about collinearity between them here.
- Genhlth and Self-Reported Health Conditions: Additional EDA checks also showed that variables such as Body Mass Index (bmi) and general health (genhlth) take on intuitive, population wide distributions which allow for strong signal to predict both active and inactive groups.
- Regional Income Patterns: To evaluate geographic trends and socioeconomic disparities, respondent income brackets were compared across states. When stratified by household income levels, the proportion of physical inactivity was greater in the lower income groups in both California and Texas, identifying income as a shared macro-level risk factor.

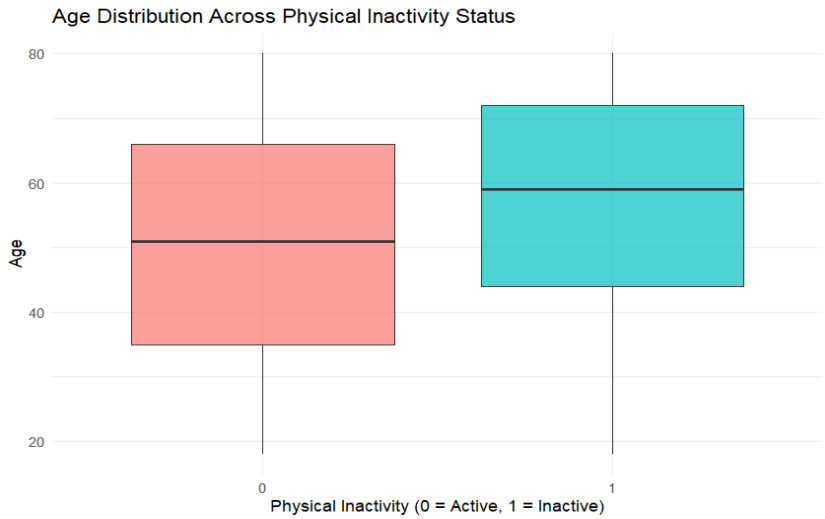



*Figure 1: Age Distribution Across Physical Inactivity Status ($n = 21,415$). Comparisons indicate a higher median age and upward shift for physically inactive adults (Group 1) relative to active adults (Group 0).*

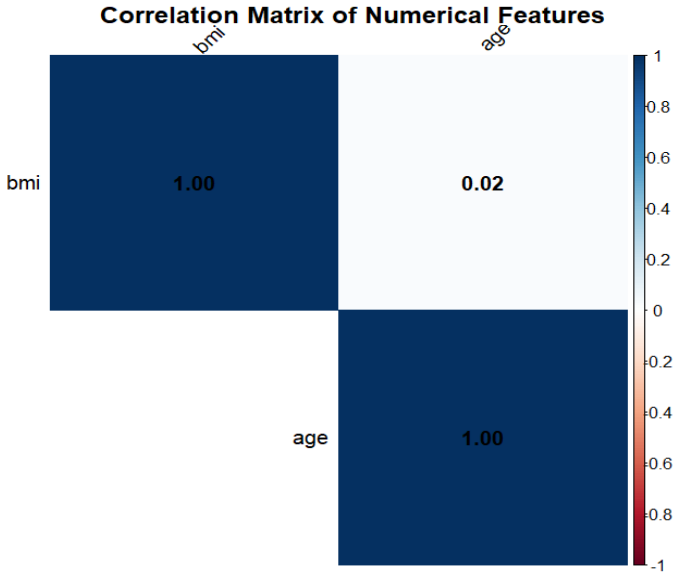



*Figure 2: Correlation Matrix of Numerical Features (bmi and age).* ***The near zero correlation coefficient (0.02) demonstrates the absence of linear collinearity between body mass index and age.***

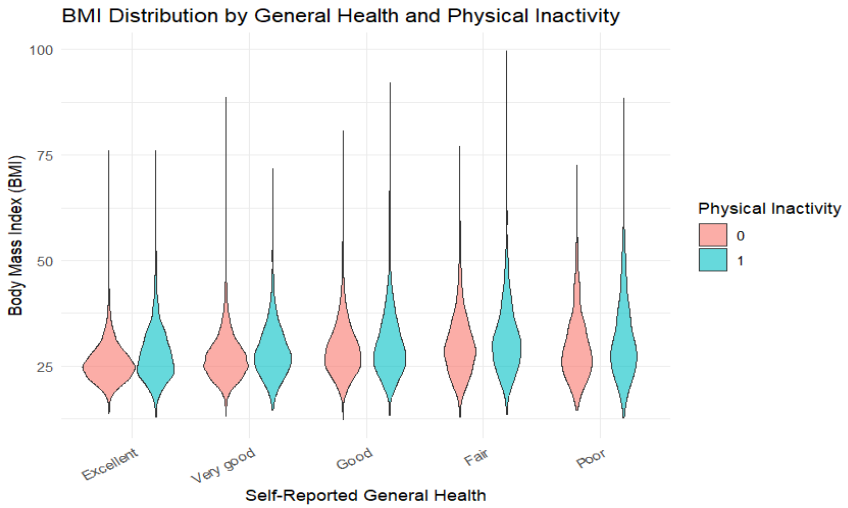



*Figure 3: BMI Distribution by Self-Reported General Health and Inactivity Status, illustrating expected population spreads across health tiers.*

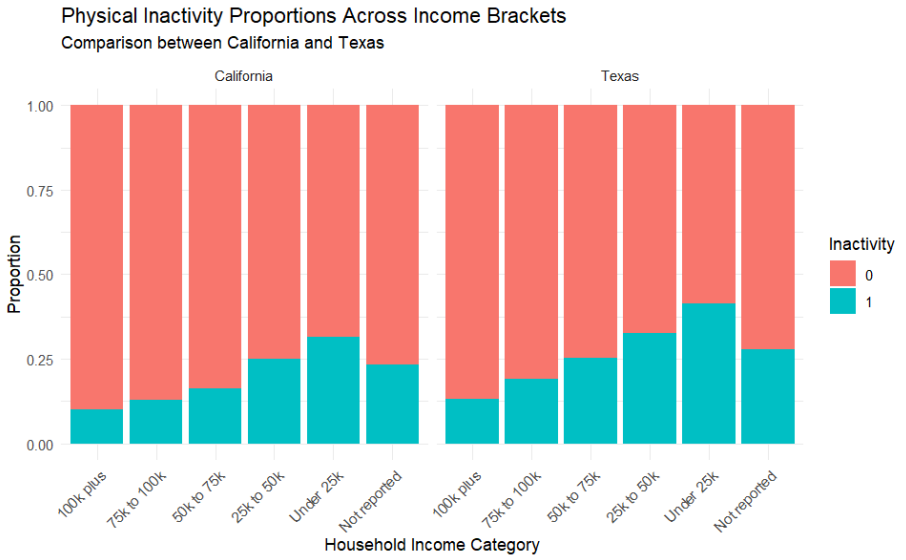



***Figure 4: When stratified by household income levels, the proportion of physical inactivity was greater in the lower income groups in both California and Texas.***

## 4.2 Phase 2: Model Training and Optimization

The data were divided into three subsets: 70% were allocated to model training, 15% to model validation, and the remaining 15% to model testing. This partitioning was performed to reduce the risk of overfitting across the four classification models defined in the study's methodological framework. Additionally, the Ridge and LASSO models employed 10fold cross-validation to determine the optimal regularization parameter using the cv.glmnet function in R.

In Ridge and LASSO regression, the regularization process aims to determine the appropriate penalty applied to the model coefficients. To accomplish this, the training data were divided into ten subsets of similar size. During each iteration, the subsets were used to fit the model,

while the remaining subset was reserved for validation. The procedure was repeated ten times so that each subset served as the validation set once.

The average validation error was then computed across the ten iterations, and the value of λ associated with the lowest average error was selected. This value was subsequently used to fit the final models. By choosing λ in this way, it helps to limit overfitting and improves the models' ability to generalize to new observations.

| Model | Cross-Validation | Selected λ |
|---|---|---|
| Ridge Regression | 10-fold | 0.008218 |
| LASSO Regression | 10-fold | 0.000409 |

***Table 2. Identified value of λ selected to train the final regularized models***

### 4.3 Phase 3 : Evaluation and Strategic Application

The performance of the four models was then evaluated and compared. The four models achieved nearly identical ROC-AUC and accuracy, indicating similar predictive performance.

| Model Strategy | Accuracy | Recall | Specificity | Precision | ROC-AUC |
|---|---|---|---|---|---|
| Logistic | 78.35% | 19.44% | 95.84% | 58.13% | **0.7431** |
| Stepwise | 78.35% | 19.44% | 95.84% | 58.13% | **0.7431** |
| Ridge Regression | 78.31% | 18.01% | 96.21% | 58.50% | **0.7437** |
| LASSO Regression | 78.35% | 19.24% | 95.90% | 58.23% | **0.7433** |

**Table 3. Classification Model Performance**

The importance of the predictors was then compared across the four models. The results, summarized in Table 4, show the relative contribution of each variable to the prediction of physical inactivity.

| Predictor | Logistic | Stepwise | Ridge | LASSO | Interpretation |
|---|---|---|---|---|---|
| Texas (vs. California) | 1.232 | 1.232 | 0.206 | 0.206 | Texas residents have higher odds of inactivity. |
| BMI | 1.031 | 1.031 | 0.029 | 0.030 | Higher BMI increases inactivity risk. |
| Age | 1.013 | 1.013 | 0.012 | 0.012 | Older respondents are slightly more likely to be inactive. |
| Female | 1.289 | 1.289 | 0.248 | 0.251 | Females show higher odds of inactivity. |
| Poor Health | 6.079 | 6.079 | 1.609 | 1.761 | Strongest predictor across all models. |
| Diabetes | 1.152 | 1.152 | 0.171 | 0.143 | Diabetes has a modest positive association. |
| Less than HS | 2.059 | 2.059 | 0.700 | 0.715 | Lower education substantially increases risk. |
| Under $25k | 1.911 | 1.911 | 0.579 | 0.620 | Lower income substantially increases risk. |

**Table 4. Most Influential predictors**

The importance of the socioeconomic predictors was also compiled and compared across the four models.

| Socioeconomic Predictor | Logistic | Stepwise | Ridge | LASSO |
|---|---|---|---|---|
| Some College | 1.409 | 1.409 | 0.321 | 0.336 |
| High School Graduate | 1.766 | 1.766 | 0.539 | 0.562 |
| Less than High School | 2.059 | 2.059 | 0.700 | 0.715 |
| $75k–100k | 1.217 | 1.217 | 0.117 | 0.159 |
| $50k–75k | 1.535 | 1.535 | 0.344 | 0.395 |
| $25k–50k | 1.917 | 1.917 | 0.571 | 0.622 |
| Under $25k | 1.911 | 1.911 | 0.579 | 0.620 |
| Income Not Reported | 1.689 | 1.689 | 0.442 | 0.493 |

**Table 5. Socioeconomic predictors**

Because the dataset is imbalanced, with approximately 23% of individuals classified as physically inactive and 77% as physically active, model performance was not evaluated using only the conventional 0.50 classification threshold. Instead, the decision threshold was calibrated by testing values from 0.10 to 0.90 in increments of 0.05. As expected, lower thresholds increased recall by identifying a larger proportion of physically inactive individuals, although this improvement came at the expense of lower specificity.

### 4.4 Results

The four models performed almost identically. Logistic Regression and Stepwise both hit 78.35% accuracy and a ROC-AUC of 0.7431, and Ridge had the best ROC-AUC, the improvement was minimal, suggesting that the regularization methods did not provide a meaningful performance advantage.

The predictors of physical inactivity were consistent across models too. Worse self-reported health, lower education and income, higher BMI, older age, being female, and living in Texas all pushed up the odds of being physically inactive. Stepwise ended up keeping every predictor, meaning none of them were dead weight.

Lastly, the standard 0.50 threshold gives strong specificity but only catches 19.4% of physically inactive people. Dropping the threshold to around 0.20 pushes recall up to 72.4%, which strikes a much better balance between flagging at-risk individuals and keeping the model's overall performance reasonable.

## 5. CONCLUSIONS

Although regional differences were observed between California and Texas, the results suggest that physical health may play a larger role in physical inactivity than socioeconomic factors alone. In particular, poor self-reported health consistently emerged as the strongest predictor across all models. This finding highlights the need for future research focused specifically on Texas to better understand which chronic conditions or health problems are most closely associated with physical inactivity. It is possible that physical inactivity is not the primary public health challenge itself, but rather a consequence of a higher burden of chronic disease or limited access to appropriate healthcare. If this is the case, public health policies aimed at reducing

physical inactivity may be more effective if they place greater emphasis on primary care and chronic disease prevention, rather than focusing exclusively on socioeconomic, demographic, or behavioral factors.

### 5.1 Lessons we have learned

This project helped us better understand several machine learning methods and how they can be applied to support public policy decisions. I also learned the importance of splitting a dataset into training, validation, and test sets to properly evaluate model performance and avoid overfitting. One aspect I found challenging was translating the concepts covered in class into working code. While the theoretical material was nice, implementing the methods in R often involved debugging and overcoming coding issues. I would like to recommend adopting an approach like MGT 6311, where theoretical concepts are taught alongside their implementation in R. I believe this would make it easier for students to connect the underlying methodology with its practical application.

## 6. APPENDIX

### CODE

```
need <- c("haven", "dplyr", "pROC")
for (p in need) {
  if (!requireNamespace(p, quietly = TRUE)) install.packages(p)
}
library(haven)
library(dplyr)
library(pROC)

setwd("/Users/jshow2211/Desktop/7406_Project")

keep <- c("_STATE", "_TOTINDA", "_BMI5", "DIABETE4",
       "EDUCA", "INCOME3", "SEXVAR", "_AGE80", "GENHLTH")

raw <- read_xpt("LLCP2022.XPT", col_select = all_of(keep))
cat("Rows read (all states):", nrow(raw), "\n")

names(raw) <- c("STATE", "TOTINDA", "BMI5", "DIABETE4",
           "EDUCA", "INCOME3", "SEXVAR", "AGE80", "GENHLTH")

raw <- raw %>% mutate(across(everything(), as.numeric))

dat <- raw %>% filter(STATE %in% c(6, 48))
```

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```
cat("Rows in CA and TX:", nrow(dat), "\n")

dat <- dat %>%
 mutate(
  inactive = case_when(TOTINDA == 1 ~ 0,    # reversed so 1 = inactive
                TOTINDA == 2 ~ 1,
                TRUE ~ NA_real_),

  state = factor(STATE, levels = c(6, 48), labels = c("California", "Texas")),

  bmi = BMI5 / 100,                  # stored with two implied decimals

  age = ifelse(AGE80 >= 18 & AGE80 <= 80, AGE80, NA),

  sex = factor(SEXVAR, levels = c(1, 2), labels = c("Male", "Female")),

  genhlth = case_when(GENHLTH %in% 1:5 ~ GENHLTH, TRUE ~ NA_real_),
  genhlth = factor(genhlth, levels = 1:5,
            labels = c("Excellent", "Very good", "Good", "Fair", "Poor")),

  diabetes = case_when(DIABETE4 == 1 ~ "Yes",
                DIABETE4 == 4 ~ "Prediabetes",
                DIABETE4 %in% c(2, 3) ~ "No",   # 2 is gestational only
                TRUE ~ NA_character_),
  diabetes = factor(diabetes, levels = c("No", "Prediabetes", "Yes")),

  educ = case_when(EDUCA %in% 1:3 ~ "Less than HS",
             EDUCA == 4 ~ "HS grad",
             EDUCA == 5 ~ "Some college",
             EDUCA == 6 ~ "College grad",
             TRUE ~ NA_character_),
  educ = factor(educ, levels = c("College grad", "Some college",
                       "HS grad", "Less than HS")),

  income = case_when(INCOME3 %in% 1:4 ~ "Under 25k",
               INCOME3 %in% 5:6 ~ "25k to 50k",
               INCOME3 == 7 ~ "50k to 75k",
               INCOME3 == 8 ~ "75k to 100k",
               INCOME3 %in% 9:11 ~ "100k plus",
               TRUE ~ NA_character_),
  income = factor(income, levels = c("100k plus", "75k to 100k", "50k to 75k",
```

```
                        "25k to 50k", "Under 25k"))
 ) %>%
 select(state, inactive, bmi, age, sex, genhlth, diabetes, educ, income)

cat("\nMissing by variable:\n")
print(colSums(is.na(dat)))
cat("\nPercent missing:\n")
print(round(100 * colSums(is.na(dat)) / nrow(dat), 2))

clean <- dat %>% filter(complete.cases(.))
cat("\nBefore complete-case filter:", nrow(dat), "\n")
cat("After: ", nrow(clean), "\n")
cat("Retained:", round(100 * nrow(clean) / nrow(dat), 1), "percent\n")

cat("\nOutcome distribution (1 = inactive):\n")
print(table(clean$inactive))
print(round(prop.table(table(clean$inactive)), 4))

cat("\nInactivity rate by state:\n")
print(clean %>% group_by(state) %>%
    summarise(n = n(), rate = round(mean(inactive), 4)))

set.seed(7406)
idx <- unlist(lapply(split(seq_len(nrow(clean)), clean$inactive),
              function(i) sample(i, size = floor(0.7 * length(i)))))

train <- clean[idx, ]
test  <- clean[-idx, ]

cat("\nTrain:", nrow(train), " Test:", nrow(test), "\n")
cat("Train inactive rate:", round(mean(train$inactive), 4), "\n")
cat("Test inactive rate: ", round(mean(test$inactive), 4), "\n")

evaluate <- function(prob, truth, model_name, cutoff = 0.5) {
 pred <- ifelse(prob > cutoff, 1, 0)
 cm <- table(Predicted = factor(pred, levels = c(0, 1)),
        Actual    = factor(truth, levels = c(0, 1)))
 tn <- cm[1, 1]; fp <- cm[2, 1]; fn <- cm[1, 2]; tp <- cm[2, 2]
 roc_obj <- roc(truth, prob, quiet = TRUE)

 data.frame(
```

```
  model     = model_name,
  accuracy   = round((tp + tn) / sum(cm), 4),
  recall     = round(tp / (tp + fn), 4),
  specificity = round(tn / (tn + fp), 4),
  precision  = round(tp / (tp + fp), 4),
  auc       = round(as.numeric(auc(roc_obj)), 4),
  stringsAsFactors = FALSE
 )
}

save(train, test, clean, evaluate, file = "brfss_clean.RData")
cat("\nSaved brfss_clean.RData\n")

library(ggplot2)
library(corrplot)
library(haven)
library(dplyr)
library(glmnet)
library(MASS)
library(caret)
library(pROC)

load("brfss_clean.RData")

# Age Distribution Across Physical Inactivity Status (Boxplot)
ggplot(clean, aes(x = factor(inactive), y = age, fill = factor(inactive))) +
 geom_boxplot(alpha = 0.7) +
 labs(
  title = "Age Distribution Across Physical Inactivity Status",
  x = "Physical Inactivity (0 = Active, 1 = Inactive)",
  y = "Age"
 ) +
 theme_minimal() +
 theme(legend.position = "none")

# BMI Distribution by General Health & Inactivity (Violin Plot)
ggplot(clean, aes(x = genhlth, y = bmi, fill = factor(inactive))) +
 geom_violin(position = position_dodge(0.8), alpha = 0.6) +
 labs(
  title = "BMI Distribution by General Health & Inactivity",
  x = "Self-Reported General Health",
```

```
   y = "Body Mass Index (BMI)",
   fill = "Inactivity"
 ) +
 theme_minimal() +
 theme(axis.text.x = element_text(angle = 30, hjust = 1))

# Correlation Matrix of Numerical Features (BMI and Age)
numeric_vars <- clean[, c("bmi", "age")]
cor_matrix <- cor(numeric_vars, use = "complete.obs")

corrplot(cor_matrix, method = "color", type = "upper",
      addCoef.col = "black",
      tl.col = "black", tl.srt = 45,
      title = "Correlation Matrix of Numerical Features",
      mar = c(0,0,2,0))



#LOGISTIC MODEL-----------------------------------------
#TRAINING PART
logistic <- glm(inactive ~ ., data = train, family = binomial)
# VALIDATION PART
probability <- predict(logistic,newdata = test, type = "response")
pred <- ifelse(probability > 0.5, 1, 0)
#Confusion matrix
confusionMatrix(factor(pred),factor(test$inactive))
#AUC
logis <- roc(test$inactive,probability)
auc(logis)

# Stepwise Selection -------------------------------------
Stepwise <- stepAIC(logistic,direction = "both", trace = FALSE)
# Prediction
probstep <- predict(Stepwise,newdata = test,type = "response")
predstep <- ifelse(probstep > 0.5, 1, 0)
#Confusion matrix
confusionMatrix(factor(predstep),factor(test$inactive))
# AUC
roc_step <- roc(test$inactive, probstep)
auc(roc_step)

x <- model.matrix(inactive ~ ., train)[, -1]
```

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```
y <- train$inactive
testx <- model.matrix(inactive ~ ., test)[, -1]

# RIDGE METHOD ------------------------------------
set.seed(1000)
ridge <- cv.glmnet(x = x, y = y, family = "binomial",alpha = 0,nfolds = 10)
#Best λ
ridge$lambda.min
#predictions
probridge <- predict(ridge, newx = testx, s = "lambda.min", type = "response")
predridge <- ifelse(as.vector(probridge) > 0.5, 1, 0)
#Confusion matrix
confusionMatrix(factor(predridge),factor(test$inactive))
rocridge <- roc(test$inactive, as.vector(probridge))
auc(rocridge)
# LASSO METHOD ------------------------------------
set.seed(1000)
lasso <- cv.glmnet(x = x,y = y,family = "binomial",alpha = 1,nfolds = 10)
#Best λ
lasso$lambda.min
#predictions
lassoprob <- predict(lasso,newx = testx,s = "lambda.min",type = "response")
lassopred <- ifelse(as.vector(lassoprob) > 0.5, 1, 0)
#Confusion matrix
confusionMatrix(factor(lassopred),factor(test$inactive))
roclasso <- roc(test$inactive, as.vector(lassoprob))
auc(roclasso)

# evaluation
results <-
rbind(evaluate(probability,test$inactive,"Logistic"),evaluate(probstep,test$inactiv
e,"Stepwise"),evaluate(as.vector(probridge),test$inactive,"Ridge"),evaluate(as.ve
ctor(lassoprob),test$inactive,"LASSO"))

#question1
summary(logistic)
exp(coef(logistic))
summary(Stepwise)
exp(coef(Stepwise))
coef(ridge, s = "lambda.min")
coef(lasso, s = "lambda.min")
```

```
#question2
coef(lasso,s="lambda.min")
summary(logistic)$coefficients
#question3
stateTexas
stateCalifornia
#question4
results
ridge$lambda.min
lasso$lambda.min
#question5
metricas <- data.frame()
secuencia <- seq(0.1,0.9,0.05)
for(c in secuencia){

 pred <- ifelse(probability>c,1,0)

 cm <- confusionMatrix(
  factor(pred),
  factor(test$inactive),
  positive="1")

 metricas <- rbind(metricas,

            data.frame(

             cutoff=c,

             Recall=cm$byClass["Sensitivity"],

             Specificity=cm$byClass["Specificity"]

            )

 )

}
metricas
```

## 7. BIBLIOGRAPHY

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